

A Defensive Short-Horizon Reversal Strategy with Order-Book Microstructure for Long-Only A-Share Portfolios

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Abstract

We propose and evaluate a long-only quantitative strategy for the Shanghai A-share market, designed to extract return predictability from a combination of price-based and order book signals. The market context shapes the design: strong short-horizon reversals, retail-driven order flow, and informative intraday book dynamics make this setting well suited to microstructure-augmented factor composites.

The approach combines two components. The first is a factor composite built from a diversified library of daily and limit order book signals, weighted by predictive stability and partially neutralized against estimated sector exposures. The second is a walk-forward Ridge regression sleeve, trained on all candidate factors without look-ahead. The key design principle is ensembling for diversification: rather than stacking correlated factors to maximize raw predictive correlation, we blend a weak but decorrelated machine learning signal into a robust base composite, seeking breadth of information as the primary source of improvement.

In-sample (≈ 1.9 years), the strategy delivers CAGR of +17.3% and a Sharpe ratio of 1.42. Out-of-sample (≈ 1 year, from a fresh capital position), it achieves CAGR of +17.6% and a Sharpe ratio of 1.00. Growth does not decay from in-sample to out-of-sample, and maximum drawdown improves, suggesting the strategy generalizes well to unseen data.

The main limitations are limited capacity, reliance on a single market regime, and idealized execution assumptions. The strategy is designed for the competition's RMB 50M mandate and has not been tested in a sustained bear market or liquidity crisis.

1 Introduction

Long-only equity strategies restrict portfolio positions to non-negative weights, and they remain the dominant operating framework in institutional asset management. Their appeal rests on both regulatory necessity (many institutional mandates prohibit short-selling) and portfolio theory: skilled active management is most efficiently expressed through conviction-weighted overweights relative to a benchmark [1, 2]. In China’s A-share market, the long-only constraint is especially binding. Securities lending is limited, short-selling mechanisms are less developed than in most major markets, and the investor base is dominated by retail participants who account for the majority of daily trading volume [3]. This produces distinctive price dynamics: pronounced short-horizon reversals driven by retail order-flow imbalances, lottery-seeking behavior that amplifies cross-sectional return dispersion, and informative intraday order book activity. Quantitative strategies in this setting must also contend with the T+1 settlement rule, transaction costs, lot-size constraints, and uneven liquidity across names.

The standard quantitative toolkit for factor evaluation in China relies heavily on the Rank Information Coefficient (Rank IC), the cross-sectional Spearman correlation between a factor and subsequent returns, and its risk-adjusted form, the IC Information Ratio (ICIR) [1]. Barra-style risk models or sector decompositions are used to neutralize systematic biases and isolate residual return predictability [4]. These tools measure statistical predictability effectively, but they assess signal quality in isolation from portfolio implementation. A factor with a high IC may still underperform net of costs if it concentrates in illiquid names, generates high turnover, or creates sector tilts that are difficult to manage under a long-only mandate [5]. For a fully-invested, long-only A-share portfolio operating under T+1 settlement, per-side commissions, stamp duty, and a minimum diversification requirement, IC maximization alone is not a sufficient design criterion. The more relevant test is end-to-end portfolio return, net of all costs, on genuinely out-of-sample data.

We ground the factor library in four return anomalies that fit this market context well. The low-volatility anomaly documents that low-idiosyncratic-volatility stocks tend to earn superior risk-adjusted returns, a pattern attributed partly to leverage constraints and partly to retail investor preference for high-beta names [6]. Short-horizon price reversals, at horizons from one to twenty days, arise from temporary liquidity imbalances and retail overreaction, and are especially pronounced in retail-dominated markets [7, 8]. The anti-lottery effect reflects the tendency for positively skewed stocks to be overpriced due to gambling-like demand, with subsequent mean-reversion as prices converge toward fundamentals [9, 10]. Finally, signed order-flow imbalance and spread dynamics from the limit order book carry short-horizon predictive content for price movements, providing information that is largely orthogonal to price-based signals [11]. These four themes offer complementary sources of return. Combining them effectively, without overfitting or excessive turnover, is the central portfolio construction challenge that motivates this study.

This paper makes four contributions. First, we construct a multi-theme factor composite through IC-IR screening and sign-stability tests [1], with a limit of two factors per economic theme to prevent any single theme from dominating the signal [12]. Second, we show that value can be added through decorrelated ensembling rather than factor stacking: a walk-forward Ridge regression sleeve [13], purged of look-ahead, improves portfolio risk-adjusted returns by contributing complementary information rather than by being a stronger predictor [14]. Third, we incorporate eleven daily factors derived from the 10-minute, 10-level limit order book, providing signals that are largely orthogonal to the price-based factor family [11]. Fourth, we layer risk management overlays calibrated to growth, risk-adjusted return, and drawdown objectives, and verify that the resulting portfolio generalizes out of sample without material performance decay.

2 Data and Trading Environment

2.1 Datasets and Descriptive Statistics

Two synchronized datasets are used, both restricted to the supplied Shanghai Stock Exchange universe. The first comprises 10-minute, 10-level limit order book (LOB) snapshots at 24 intraday time points (09:40-15:00), providing ten levels of bid and ask prices and volumes per asset per snapshot. The second is a daily OHLCV dataset containing open, high, low, and closing prices, trading volume and amount, a price adjustment factor, and a 09:30-09:35 volume-weighted average price (VWAP) used as the buy execution reference. All prices are adjusted as $\text{adj_close} = \text{close} \times \text{adj_factor}$ prior to any return or factor computation.

The in-sample (IS) panel spans 484 trading days (D001-D484) and covers 2,270 unique assets, yielding 1,060,121 asset-day observations. The combined IS and out-of-sample (OOS) panel spans 726 trading days across 2,306 assets. The first 80 trading days serve as a factor warm-up window; active portfolio positioning begins at D080. Approximately 11.2% of assets satisfy the eligibility criteria on a typical IS day, rising to approximately 12.7% in the OOS period.

2.2 Time Split and Leakage Policy

The dataset is partitioned into an IS period (D001-D484) used for all factor screening, weighting, sector estimation, and model training, and an OOS period (D485-D726) reserved exclusively for inference. No IS-fitted artifact is re-estimated or adjusted on any OOS observation; the OOS portfolio is driven entirely by parameters fixed during the IS phase.

2.3 Eligibility Universe

On each trading day, stocks are eligible for purchase only if they pass four filters simultaneously. Prices at all execution points (open, close, and 09:30-09:35 VWAP) must be valid and strictly positive. The 20-day median traded amount must be at least RMB 300,000. The stock must not be at a prior limit-up state. And the stock must have recorded positive trading volume over the trailing three days, indicating it has not been recently suspended. This screen concentrates the portfolio in liquid, tradable names and reduces the risk of unfillable orders.

2.4 Execution Rules

The backtest engine mirrors the official competition rules exactly. Settlement follows the T+1 rule, so stocks purchased on day t cannot be sold before day $t + 1$. On each trading day, sell orders are processed before buy orders. Buys execute at the 09:30-09:35 VWAP; sells execute at the opening price under the committed *sell-at-open* mode. Commission is $\max(\text{turnover} \times 1 \text{ bp}, \text{RMB } 5)$ on both sides; stamp duty is 5 bp on sells only. The minimum trade unit is 100 shares with shrink-to-fit lot rounding; there is no leverage, and any order lacking sufficient cash or a valid price receives no fill. Net asset value (NAV) is marked at the daily closing price.

3 Factor Construction

All factors are computed on the full price history so that day- t look-backs are correct, but every factor is lagged by one trading day so that a day- t investment decision uses only information available through the close of day $t - 1$. Every fitted artifact is estimated on IS data only. We first describe the factor families, then normalization, and finally the construction of the traded composite signal. Complete factor definitions are reported in Appendix C.

Notation: for asset i and day t , let $r_{i,t}$ denote the adjusted close-to-close log return and $P_{i,t}^{\text{adj}}$ the adjusted closing price.

3.1 Daily Factor Families

The daily factor library comprises 16 factors spanning five economic themes, each signed so that a higher value predicts higher expected forward return. The short-horizon reversal family (`REV_h` for horizons $h \in \{1, 3, 5, 10, 20\}$) captures the well-documented mean-reversion tendency of daily returns at short horizons [7]. The low-volatility and calm-range family (`LOWVOL`, `RANGE_PCT`, `RANGE_TREND`) exploits the low-volatility anomaly [6]. The intraday microstructure family (`OVNT`, `MORN_REV`, `INTRA_MOM`, `CLOSE_STRENGTH`) captures overnight and intraday price formation dynamics. The anti-lottery family (`SKEW_60`) measures negative co-skewness preference [9, 10]. Finally, the liquidity and turnover family (`ILLIQ`, `AMT_GROWTH_5`) uses the Amihud illiquidity ratio [15] and turnover acceleration as proxies for liquidity conditions. An intermediate momentum factor (`MOM_60_20`) is also included as a control, using the standard skip-month construction.

3.2 LOB Factor Family

We aggregate the 10-minute, 10-level order book to one daily observation per asset. For each intraday snapshot, the level-1 order-flow imbalance and the depth-weighted imbalance across L levels are defined as:

$$\text{OFI} = \frac{\text{bidvol}_1 - \text{askvol}_1}{\text{bidvol}_1 + \text{askvol}_1}, \quad \text{DWI}_L = \sum_{k=1}^L \frac{1}{k} \frac{\text{bidvol}_k - \text{askvol}_k}{\text{bidvol}_k + \text{askvol}_k}, \quad (1)$$

with relative spread $\text{spr} = (\text{ask}_1 - \text{bid}_1)/\text{mid}$ and $\text{mid} = (\text{ask}_1 + \text{bid}_1)/2$. We construct eleven daily factors as intraday aggregates of these primitives, capturing order-flow timing (`OFI_LAST4`, `OFI_TREND`), spread dynamics (`SPR_D`, `SPR_RANGE`), depth-weighted imbalance (`DWI_D`, `DWI10_D`, `DWI_AFT_MORN`), intraday imbalance dispersion (`OFI_RANGE`), mid-price volatility (`MIDPRICE_VOL`), and book tightness (`BOOK_TIGHTNESS`). Detailed definitions appear in Appendix C. Because single-day snapshots are noisy, all LOB factors are exponentially smoothed with a half-life of three days before use [11].

3.3 Cross-Sectional Normalization

We transform every raw factor to a daily cross-sectional z -score. On each day the factor is first winsorized at the 1st and 99th percentiles and then standardized:

$$z_{i,t} = \frac{\tilde{f}_{i,t} - \mu_t(\tilde{f})}{\sigma_t(\tilde{f})}, \quad (2)$$

where $\tilde{f}$ is the winsorized factor value and μ_t , σ_t are the cross-sectional mean and standard deviation computed within day t . This procedure uses only same-day information and introduces no temporal leakage, yielding a library of 27 standardized factors $\{z_c\}$.

3.4 BaseAlpha: IC-IR-Weighted, Family-Diversified Composite

Following the IC-IR weighting framework of Grinold and Kahn [1], predictive quality is measured by the daily rank information coefficient - the cross-sectional Spearman correlation between a factor and the 10-day forward return rank - summarized by its mean and stability:

$$\text{IC} = \overline{\text{IC}_t}, \quad \text{ICIR} = \frac{\overline{\text{IC}_t}}{\text{std}(\text{IC}_t)}. \quad (3)$$

A factor is retained only if it clears all four screening gates: $|\text{IC}| \geq 0.010$; sign-stability across both IS halves (each half-period IC at least 20% of the full-sample magnitude and of the same

sign); $|\text{ICIR}| \geq 0.12$; and pairwise rank-correlation to any anchor factor ≤ 0.65 to remove redundant signals. Among survivors, each factor family is capped at two members to enforce breadth [12], and the top eight by $|\text{ICIR}|$ are retained. Each is assigned a weight proportional to its signed IC-IR:

$$w_c = \frac{\text{sign}(\text{IC}_c) |\text{ICIR}_c|}{\sum_{c'} |\text{ICIR}_{c'}|}, \quad \text{BaseAlpha}_{i,t} = \sum_c w_c z_{c,i,t}. \quad (4)$$

The eight selected factors and their IC-IR weights are reported in Table 1. The composite is dominated by defensive low-risk factors (**LOWVOL**, **RANGE_PCT**), anti-lottery skewness (**SKEW_60**), and multi-horizon reversal (**REV_20**, **REV_10**, **OVNT**), with two order-book factors (**DWI_AFT_MORN**, **SPR_D**) providing decorrelated microstructure signals. Predictive statistics for each signal stage are summarized in Table 2.

Table 1: IC-IR weights of the eight selected factors.

Factor	Weight	Factor	Weight
z_LOWVOL	+0.174	z_DWI_AFT_MORN	−0.118
z_RANGE_PCT	+0.156	z_SPR_D	−0.109
z_SKEW_60	+0.155	z_REV_10	+0.074
z_REV_20	+0.141	z_OVNT	−0.073

3.5 Sector Residualization

Because GICS-style sector labels are not provided, we estimate statistical sectors from return co-movement: principal-component reduction (20 components) of the D080-D240 daily return matrix followed by k -means clustering into 10 groups. The sector map is frozen from this early IS window [4]. The within-sector mean of the alpha is then partially removed:

$$\text{BaseAlpha_res}_{i,t} = \text{BaseAlpha}_{i,t} - \lambda \overline{\text{BaseAlpha}}_{s(i),t}, \quad \lambda = 0.5, \quad (5)$$

where $s(i)$ denotes the cluster of asset i . Partial (rather than full) neutralization preserves some sector tilt while reducing concentration. As shown in Table 2, ICIR improves substantially despite a small reduction in raw IC.

3.6 Ridge Ensemble Sleeve

In parallel, we estimate a walk-forward ridge regression of the 5-day forward rank on all 27 standardized factors [13]:

$$\hat{\beta}_\tau = \arg \min_{\beta} \sum_{(i,t) \in \mathcal{T}_\tau} (\text{rank}_{i,t}^{(5)} - z_{i,t}^\top \beta)^2 + \alpha \|\beta\|_2^2, \quad \alpha = 10. \quad (6)$$

The model is retrained every 30 days. Each training set $\mathcal{T}_\tau$ uses only observations dated earlier than (prediction day -12); the 12-day purge gap exceeds the 5-day target horizon, ensuring that no training label overlaps the prediction date [12]. Predictions are standardized cross-sectionally to give **z_ridge5**. Standalone, the sleeve is comparatively weak as a predictor, but sufficiently decorrelated from the base composite to contribute ensemble diversification value (see Table 2).

3.7 Final Alpha

The traded signal blends the two standardized components:

$$\text{Final Alpha}_{i,t} = 0.8 \cdot z(\text{BaseAlpha_res})_{i,t} + 0.2 \cdot z(\text{ridge5})_{i,t}. \quad (7)$$

As shown in Table 2, Final Alpha achieves a higher IC than either component alone, suggesting the two signals carry complementary information.

4 Portfolio Construction and Execution

We convert the signal into orders via a single deterministic engine applied identically in-sample and out-of-sample.

4.1 Selection and Sizing

On each rebalance day, the portfolio targets the top $N = 12$ eligible names by Final Alpha rank, subject to a hard minimum of 10 holdings (the competition floor) and a per-name position cap of 9%. To reduce unnecessary turnover and associated cost drag, incumbent holdings that remain within the top-50 ranked names are retained without triggering a rebalance, and the portfolio is rebalanced only once every 10 trading days. Target weights blend inverse-volatility and alpha-rank allocations in equal proportions,

$$w_i \propto 0.5 \cdot \frac{1/\sigma_i}{\sum_j 1/\sigma_j} + 0.5 \cdot \frac{\text{rank}(\alpha_i)}{\sum_j \text{rank}(\alpha_j)}, \quad (8)$$

then capped at 9% and renormalized. Inverse-volatility dampens the contribution of the most volatile holdings, while alpha-rank tilts toward higher-conviction names; the equal blend exploits the diversification benefits of combining the two approaches [14].

4.2 Risk Overlays

We apply three risk overlays on top of the trading signal, each targeting a distinct dimension of portfolio performance. Volatility targeting scales gross exposure toward an 18% annualized target using the trailing 20-day realized portfolio volatility, with adjustments bounded within $[0.90, 1.10]$. A breadth gross-up overlay raises the upper exposure bound to 1.20 when more than 55% of eligible names show a positive 20-day return, allowing the portfolio to lean into broad up-market conditions. A drawdown breaker cuts exposure to a floor of 0.50 and gradually restores it over three days whenever the portfolio falls more than 5% below its recent peak.

4.3 Order Generation and Leakage Controls

On each rebalance day the engine first liquidates names leaving the target set (full exits at the opening price), frees the resulting cash, and then issues buy orders sequentially against the remaining cash at the 09:30-09:35 VWAP, respecting lot rounding and the no-leverage rule. The pipeline is designed to prevent look-ahead at every stage. Factors are lagged by one day; BaseAlpha weights are estimated on in-sample data only, with the forward-return target masked to ensure no overlap with the out-of-sample period; the sector map is frozen on the early IS window; and the Ridge sleeve is trained on historical data only, with a 12-day purge gap [12]. Forward returns are used only as training targets and IC diagnostics, never as inputs to a live trading decision.

5 Empirical Analysis

5.1 Predictive Strength and Factor Correlation

Table 2 reports the in-sample IC and ICIR for each signal component. Sector residualization trades a small reduction in raw IC for a large gain in stability (ICIR from 0.674 to 0.815). The ensemble achieves a higher IC than either component alone (0.0980 versus 0.1026 and 0.0945 individually), confirming complementary information content between the IC-IR composite and the Ridge sleeve.

Table 2: In-sample predictive strength of each signal component.

Signal	IC	ICIR
BaseAlpha (raw)	0.1026	0.674
BaseAlpha (sector-residualized)	0.0947	0.815
Ridge sleeve (z_ridge5)	0.0945	0.619
Final Alpha (ensemble)	0.0980	0.795

Figure 1 displays the pairwise rank-correlation matrix of the eight selected factors. All off-diagonal correlations lie below the 0.65 screening threshold, consistent with the diversification objective of the factor selection procedure. The strongest positive correlation (≈ 0.6) exists between the two reversal factors (REV_10 and REV_20). The microstructure factors (DWI_AFT_MORN, SPR_D) show near-zero correlation with price-based factors, suggesting that the LOB signals carry information largely distinct from the daily factor family.

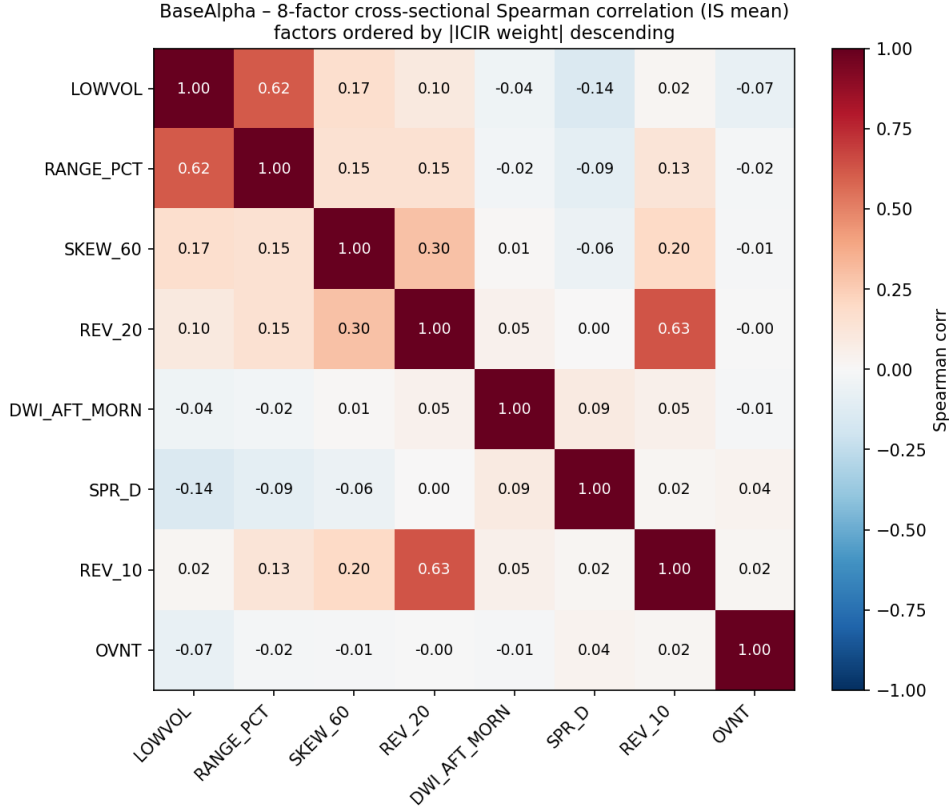


Figure 1: Pairwise rank-correlation matrix of the eight IC-IR-selected factors (in-sample, D080-D484). All off-diagonal entries lie below the 0.65 redundancy threshold, confirming breadth diversification.

5.2 Signal Decomposition: The Ensemble Effect

To isolate the contribution of each component, we apply the same portfolio engine to each signal in turn. Table 3 and Figure 2 report the results. The Ridge sleeve is a poor standalone strategy, yet allocating 20% of the signal weight to it lifts both Sharpe and CAGR substantially while leaving drawdown essentially unchanged. The gain appears to reflect a diversification effect rather than the substitution of a stronger predictor [14]. This constitutes the main empirical result of the study.

Table 3: Signal decomposition: identical portfolio engine applied in-sample (D080-D484).

Signal	IC	ICIR	CAGR	Sharpe	MaxDD	Calmar	Turnover	Fitness	Final NAV
BaseAlpha (sector-resid)	0.0947	0.815	+12.41%	1.00	-12.26%	1.01	10.51×	0.11	62.59M
Ridge sleeve	0.0945	0.619	+1.49%	0.18	-24.53%	0.06	11.66×	0.01	51.44M
Final Alpha (0.8/0.2)	0.0980	0.795	+17.26%	1.42	-12.32%	1.40	10.96×	0.18	67.88M



Figure 2: In-sample signal decomposition: equity curves and drawdowns for BaseAlpha, the Ridge sleeve, and the combined Final Alpha (D080-D484).

5.3 In-Sample Portfolio Performance



Figure 3: In-sample equity curve and drawdown of the Final Alpha strategy (D080-D484), RMB 50M start.

Over the in-sample period the strategy compounds RMB 50M to **RMB 67.88M** (+35.8% total), delivering **CAGR +17.26%**, **Sharpe 1.42**, **MaxDD -12.32%**, **Calmar 1.40**. The drawdown profile remains within a -12.3% envelope, consistent with the design intent of the volatility-targeting and drawdown-breaker overlays.

5.4 Out-of-Sample Generalization

The OOS portfolio begins with a fresh RMB 50M on D485 and trades through D726 using exclusively IS-fitted artifacts. Table 4 summarizes performance across both evaluation windows.

Table 4: In-sample and out-of-sample portfolio performance summary.

Window	Period	Days	CAGR	Sharpe	MaxDD	Calmar	Turnover	Fitness	Final NAV
In-sample	D080-D484	≈ 484	+17.26%	1.42	-12.32%	1.40	10.96×	0.18	67.88M
Out-of-sample	D485-D726	242	+17.64%	1.00	-10.18%	1.73	11.20×	0.13	58.44M

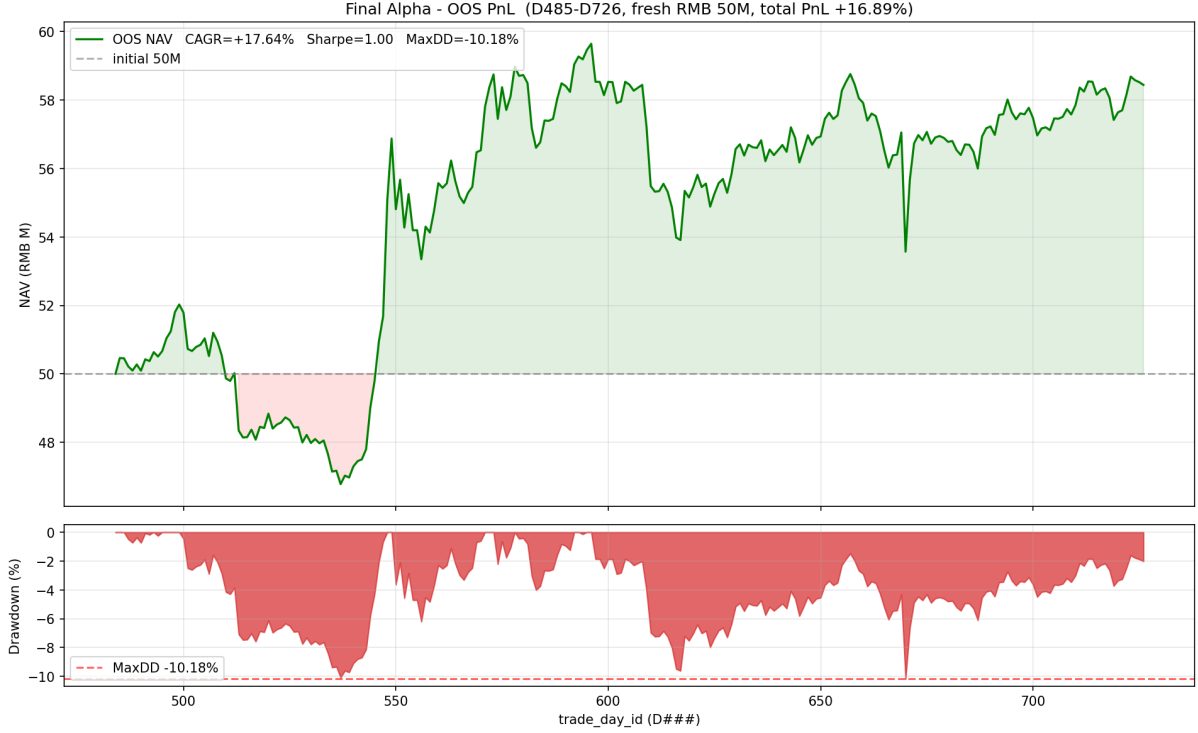


Figure 4: Out-of-sample equity curve and drawdown (D485-D726), fresh RMB 50M cold start; total PnL +16.89%.

Out-of-sample the strategy grows RMB 50M to **RMB 58.44M** (+16.89% total) with **CAGR** +17.64%, **Sharpe 1.00**, **MaxDD** −10.18%. Three patterns are noteworthy. OOS CAGR (+17.64%) modestly exceeds its in-sample counterpart, suggesting the growth signal is not over-fitted. The OOS maximum drawdown (−10.18%) is shallower than in-sample (−12.32%), lifting OOS Calmar to 1.73 and indicating that the risk overlays continue to operate as intended. OOS Sharpe declines from 1.42 to 1.00, a reduction consistent with the shorter evaluation window (approximately one year) and the greater uncertainty inherent in unseen market conditions.

5.5 Submission Compliance and Mapping to Composite Score

The strategy emits an instruction file of 181 order rows (121 buys, 60 sells) across 23 rebalancing days (D485-D705) under sell-at-open mode. An independent re-pricing of the CSV from a RMB 50M cold start under the literal Section-4 rules reproduces the strategy NAV to a maximum relative error of 3.2×10^{-16} and confirms a minimum of 10 holdings on every OOS day. All fifteen format and rule checks pass.

The risk overlays are each calibrated to one dimension of the scoring structure, and the realized OOS results are competitive on all three axes simultaneously.

6 Innovation and Limitations

6.1 Innovation

The main innovation is the use of ensembling as a source of diversification rather than a means of amplifying any single predictor. Instead of stacking correlated factors to maximize raw IC, we add value by blending a decorrelated and individually weak ML sleeve into a robust composite. Section 5.1 shows that the improvement in Sharpe is attributable to breadth of information, which tends to be more durable than relying on a stronger single signal [14]. The IC-IR screening

procedure, with its sign-consistency requirements, redundancy gates, and two-per-family caps, produces a composite resistant to single-theme failure, consistent with the importance of breadth-based diversification in empirical asset pricing research [12, 1]. The integration of eleven limit order book factors injects signals that are largely orthogonal to the price-based family, motivated by the evidence on order-flow predictability in microstructure research [11]. We calibrate the risk overlays to growth, drawdown, and volatility objectives, translating the scoring structure into concrete portfolio decisions.

6.2 Limitations and Model Risk

Several limitations deserve acknowledgment. The statistical sector model is estimated from an early in-sample window and frozen for the entire evaluation period. The resulting clusters approximate, but do not equal, true industry groupings, and may drift over longer horizons [4]. The concentrated portfolio raises idiosyncratic risk; position caps and the minimum holding floor reduce, but do not eliminate, this exposure. The execution model assumes fills at the official reference prices, without market impact or partial-fill slippage; at meaningful scale, realized execution may diverge from these assumptions [5]. Relatedly, the strategy is designed for the competition’s RMB 50M mandate and has not been tested for large-AUM deployment. The approximately one-year OOS period yields Sharpe estimates with wide confidence bounds; point estimates should be read as indicative rather than precise. Finally, the data cover a single market regime. How the strategy would behave in a sustained bear market or liquidity crisis is untested beyond the drawdown breaker’s design parameters.

7 Conclusion

This study presents a long-only A-share strategy whose traded signal, *Final Alpha*, combines two components: a factor composite built from low-volatility, anti-lottery, reversal, and order-book signals, partially neutralized against estimated sector exposures; and a walk-forward Ridge regression sleeve trained with strict look-ahead controls. The combined signal drives a concentrated, volatility-targeted portfolio governed by breadth and drawdown overlays.

In-sample, the strategy returns CAGR +17.26% at Sharpe 1.42 with a −12.32% maximum drawdown. Out-of-sample, from a fresh capital start, it achieves CAGR +17.64% at Sharpe 1.00 with a shallower −10.18% drawdown. The absence of CAGR decay and the improvement in drawdown out-of-sample are consistent with a design that generalizes rather than overfits. The key design principle is that ensembling for diversification, blending a weak but decorrelated ML signal into a robust composite, improves the portfolio’s risk-adjusted profile. This reflects breadth of information rather than a stronger single predictor, and tends to be more robust to changing market conditions. Factor screening, sector neutralization, and risk overlays translate this principle into a portfolio that performs consistently across growth, risk-adjusted return, and drawdown dimensions.

Natural extensions include a regime-conditional blend weight between the composite and ML sleeves, a covariance-aware risk model with shrinkage to replace inverse-volatility sizing, a capacity-aware impact model to test scalability, and a nonlinear gradient-boosted sleeve under the same strict purge discipline to probe whether further decorrelated signal can be harvested from the available factor library.

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A Strategy Configuration

Table 5: Complete strategy parameter configuration.

Group	Parameter	Value
Universe	Min 20-day median amount	RMB 300,000
Alpha	Factor library	16 daily + 11 LOB = 27
Alpha	BaseAlpha size / family cap	top 8 / 2 per family
Alpha	Sector model	PCA-20 + KMeans-10 (D080-D240)
Alpha	Sector residualization λ	0.50
Alpha	Ridge α / horizon / retrain / purge	10 / 5d / 30d / 12d
Alpha	Ensemble weight (Ridge)	0.20
Portfolio	Target names / min holdings / name cap	12 / 10 / 9%
Portfolio	Sizing	0.5 inverse-vol + 0.5 alpha-rank
Portfolio	Sticky top- K / rebalance	50 / every 10 days
Risk	Annual vol target / scale band	18% / [0.90, 1.10]
Risk	Breadth gross-up (threshold / ceiling)	0.55 / 1.20
Risk	Drawdown breaker (trigger / floor / ramp)	-5% / 0.50 / 3 days
Execution	Buy / sell price	09:30-09:35 VWAP / open
Execution	Commission / stamp / lot	1bp (min RMB5) / 5bp / 100
Capital	Initial (each of IS, OOS)	RMB 50,000,000

All figures and quantitative results in this report are produced by the accompanying reproducible pipeline: `strategy_final.ipynb` (in-sample), `strategy_signal_breakdown_IS.ipynb` (signal decomposition), and `strategy_oos_submission.ipynb` (out-of-sample submission and official-rules replay).

B Performance Metric Definitions

All portfolio metrics are computed from the daily NAV series V_t (marked at the close, net of all transaction costs) with daily returns $R_t = V_t/V_{t-1} - 1$ over n trading days, a zero risk-free rate, and annualization factor $T_{\text{yr}} = 252$.

CAGR. The Compound Annual Growth Rate is the annualized geometric mean rate of portfolio growth, expressing cumulative performance as an equivalent constant annual return independent of the length of the evaluation period [16].

$$\text{CAGR} = \left(\frac{V_n}{V_0} \right)^{T_{\text{yr}}/n} - 1.$$

Sharpe ratio. The Sharpe ratio measures the excess return earned per unit of total return volatility and is the standard risk-adjusted performance metric in asset management [17, 18]. Under the assumption of a zero risk-free rate it reduces to the ratio of mean daily return to its standard deviation, annualized by $\sqrt{T_{\text{yr}}}$.

$$\text{Sharpe} = \frac{\overline{R_t}}{\sigma(R_t)} \sqrt{T_{\text{yr}}}.$$

Maximum drawdown. Maximum drawdown quantifies the largest peak-to-trough decline in portfolio value over the evaluation period, capturing the worst cumulative loss experienced by an investor who entered the strategy at its historical peak [19].

$$\text{MaxDD} = \min_t \left(\frac{V_t}{\max_{s \leq t} V_s} - 1 \right).$$

Calmar ratio. The Calmar ratio relates annualized return to the absolute magnitude of maximum drawdown, providing a return-per-unit-of-tail-loss measure widely used in managed futures and hedge fund performance evaluation [20].

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{MaxDD}|}.$$

Turnover. Portfolio turnover measures the average fraction of the portfolio reallocated each trading day; the annualized figure is the standard cost-sensitivity diagnostic in practical factor investing, since higher turnover magnifies transaction cost drag [1].

$$\text{turnover}_t = \sum_i |w_{i,t} - w_{i,t-1}|, \quad \text{Turnover} = T_{\text{yr}} \cdot \overline{\text{turnover}_t}.$$

Fitness. The Fitness metric is adapted from the WorldQuant-style performance measure. It balances risk-adjusted return against trading intensity, rewarding a high Sharpe ratio while penalizing strategies that achieve it through excessive turnover. The $\max(\text{Turnover}, 0.125)$ floor prevents near-zero turnover from artificially inflating the score.

$$\text{Fitness} = \text{Sharpe} \times \sqrt{\frac{|\text{Return}|}{\max(\text{Turnover}, 0.125)}}.$$

C Factor Definitions

Daily factor library. Factors are signed so that a higher value predicts a higher expected forward return.

Table 6: Daily factor library (16 factors).

Factor	Definition (sign as used)	Economic motive
REV_h, $h \in \{1, 3, 5, 10, 20\}$	$-\left(P_{i,t}^{\text{adj}}/P_{i,t-h}^{\text{adj}} - 1\right)$	Short-horizon reversal
MOM_60_20	$P_{i,t-20}^{\text{adj}}/P_{i,t-60}^{\text{adj}} - 1$	Intermediate momentum
LOWVOL	$-\text{std}(r_i, 20)$	Low-volatility anomaly
ILLIQ	$-\text{mean}(r_i /\text{amount}_i, 20)$	Amihud illiquidity
OVNT	$-(\text{open}_{i,t}/\text{close}_{i,t-1} - 1)$	Overnight reversal
MORN_REV	$-(\text{vwap}_{0930-0935}/\text{open} - 1)$	Opening-auction reversal
INTRA_MOM	$\text{close}/\text{vwap}_{0930-0935} - 1$	Intraday momentum
RANGE_PCT	$-((\text{high} - \text{low})/\text{close})$	Calm-range preference
CLOSE_STRENGTH	$-((\text{close} - \text{low})/(\text{high} - \text{low}))$	Close-strength reversal
RANGE_TREND	$\text{mean}(\text{RANGE_PCT}, 5)$ $\text{mean}(\text{RANGE_PCT}, 20)$	– Range regime shift
SKEW_60	$-\widehat{\text{skew}}(r_i, 60)$	Anti-lottery
AMT_GROWTH_5	$\text{amount}_{i,t}/\text{amount}_{i,t-5} - 1$	Turnover acceleration

LOB factor family. The 10-minute, 10-level order book provides 24 intraday snapshots per asset-day, indexed $s = 1, \dots, 24$. From each snapshot we compute the level-1 order-flow imbalance OFI_s , the depth-weighted imbalances $\text{DWI}_{5,s}$ and $\text{DWI}_{10,s}$, the relative spread spr_s , the mid-price mid_s , and the book-tightness ratio, using the primitives of Section 3.2:

$$\text{OFI}_s = \frac{\text{bidvol}_1 - \text{askvol}_1}{\text{bidvol}_1 + \text{askvol}_1}, \quad \text{DWI}_{L,s} = \sum_{k=1}^L \frac{1}{k} \frac{\text{bidvol}_k - \text{askvol}_k}{\text{bidvol}_k + \text{askvol}_k},$$

$$\text{spr}_s = \frac{\text{ask}_1 - \text{bid}_1}{\text{mid}_s}, \quad \text{mid}_s = \frac{\text{ask}_1 + \text{bid}_1}{2}, \quad \text{tight}_s = \frac{\text{ask}_5 - \text{bid}_5}{\text{ask}_1 - \text{bid}_1}.$$

The eleven daily LOB factors are intraday aggregates of these primitives (Table 7). Statistics (mean_s , std_s , max_s , min_s) are taken over the snapshots s of the same trading day; “first 4” denotes $s \leq 4$, “last 4” denotes $s \geq 21$, and “morning”/“afternoon” denote $s \leq 12$ / $s > 12$. Like the daily factors, each LOB factor is signed so that a higher value predicts a higher expected forward return, is lagged one trading day, and is exponentially smoothed with a three-day half-life before cross-sectional normalization.

Table 7: LOB factor family (11 factors), aggregated daily from the 10-minute book.

Factor	Definition (sign as used)	Economic motive
OFI_D	$\text{mean}_s(\text{OFI}_s)$	Average level-1 buy pressure
DWI_D	$\text{mean}_s(\text{DWI}_{5,s})$	Depth-weighted imbalance (5 levels)
DWI10_D	$\text{mean}_s(\text{DWI}_{10,s})$	Deep-book imbalance (10 levels)
SPR_D	$-\text{mean}_s(\text{spr}_s)$	Tight-spread (liquidity) preference
OFI_LAST4	$\text{mean}_s(\text{OFI}_s \mid s \geq 21)$	End-of-day order-flow pressure
OFI_TREND	$\text{mean}_s(\text{OFI}_s \mid s \geq 21) - \text{mean}_s(\text{OFI}_s \mid s \leq 4)$	Intraday order-flow drift
OFI_RANGE	$\text{max}_s(\text{OFI}_s) - \text{min}_s(\text{OFI}_s)$	Order-flow dispersion / instability
MIDPRICE_VOL	$\text{std}_s(\text{mid}_s) / \text{mean}_s(\text{mid}_s)$	Intraday mid-price volatility
BOOK_TIGHTNESS	$\text{mean}_s(\text{tight}_s)$	Depth concentration near the touch
DWI_AFT_MORN	$\text{mean}_s(\text{DWI}_{5,s} \mid s > 12) - \text{mean}_s(\text{DWI}_{5,s} \mid s \leq 12)$	Afternoon vs. morning imbalance shift
SPR_RANGE	$\text{max}_s(\text{spr}_s) - \text{min}_s(\text{spr}_s)$	Intraday spread instability

Volume denominators are floored at one share and price denominators at 10^{-6} to guard against degenerate snapshots, exactly as implemented in the notebook pipeline.